

Department of Mechanical Engineering
B.Tech Project Presentation on

VOLLEY: Controlled CubeSat Deployment from a Hosted Orbital Platform

Completed Gen5 system-level computational evaluation



Presented By

Adityavardhan Mishra · Pratham Chawla

PRN 23070125054 · PRN 23070125029

Guide: Vikas Gulia

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SYMBIOSIS INSTITUTE OF TECHNOLOGY PUNE

Presentation route

OUTLINE

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Which missions might value orbital choice?

INTRODUCTION

Repeat fleets

Configurable smallsat families and replenishment exist. That context does not establish demand for VOLLEY.

Delivery buyer

A carrier, rideshare integrator or fleet operator might need several distinct release states from one host.

Gen5 test

Compare commanded release with springs, host manoeuvres, transport and onboard propulsion on the same mission.

No customer is confirmed; modeled Gen5 3U mass fails its preset criterion.

Existing methods solve different parts of the problem

LITERATURE REVIEW

Approach	Strength	Limit for this thesis
Spring deployer	Flight heritage; compact; simple	Limited tailored release speed
Spring release / drag	Relative state or drag can create phasing	Timing alone, with no relative state change, cannot
Orbital transfer vehicle	Larger orbit changes	Greater bus, mass and integration burden
Electromagnetic research	Commandable acceleration studied	Armature, high-g or maturity constraints vary

A narrower, testable contribution

RESEARCH GAP

Commanded release for an unmodified CubeSat at moderate acceleration, with the full system counted.

Contribution examined

Gen5 reusable-sled linear motor, 3U payload, magazine, pulse power, arrest and orbital utility.

Boundary

Electromagnetic CubeSat deployment exists in prior work. Payload qualification and physical performance remain unproved.

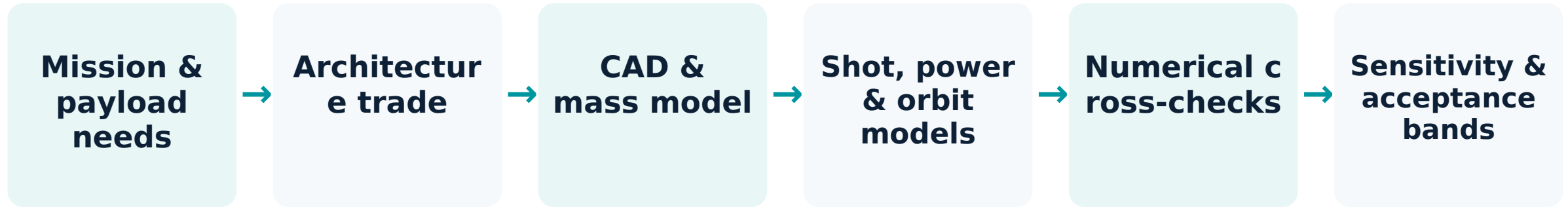
The analytical research objectives have outcomes

OBJECTIVES

Research objective	Final thesis outcome
Design a commandable 3U system	Gen5 CAD and complete subsystem architecture documented
Calculate shot and control performance	Rated shot, power and dispersion models completed
Evaluate orbital utility	Predicted energy and lifetime benefit for stated cases
Count full installed burden	Mass rollup completed; 3U parity criterion failed
Test credibility of key claims	Numerical cross-checks and limits recorded

A completed analysis chain from mission to decision

RESEARCH METHODOLOGY



Models were challenged and revised; hardware performance remains outside this thesis's evidence.

Numerical checks strengthened the final baseline

RESEARCH METHODOLOGY

Question	Method / cross-check	Evidence limit
Field and thrust	Analytic winding; magpylib; FEM	No measured thrust constant
Shot and power	Dynamics; Monte Carlo; ngspice	Assumed components and interfaces
Structure and flow	CAD; CalculiX; OpenFOAM	No payload qualification
Orbital change	Orbit model; Cowell / GMAT cases	Lifetime depends on atmosphere

Project evolution: documented decisions and analysis

TIMELINE

2023

Hosted concept Free-flyer concept reframed around a host platform

Mid-2025*

LSM direction Coilgun study gives way to a linear synchronous motor

2025-26*

CAD generations Geometry and mass models mature through Gen5

Jul-Sep 2026

Gen5 evidence Cross-checks, defect corrections and thesis record

Oct 2026

Final review Gen5 results, failed criterion and thesis record presented

* Approximate historical milestone; not a documented semester deadline.

2026 findings were incorporated in the final baseline

TIMELINE

Jul

CAD-derived sled mass replaced an optimistic parametric estimate

Aug

Depth-resolved thrust calculation lowered the rated speed

Aug

Enclosure buildup exposed the full 3U mass penalty

2026

Independent orbit checks rejected a solar-invariance claim

Final

Results and failures retained in the submitted thesis record

The current review package is assembled and traceable

WORK PROGRESS

Design delivered

Gen5 CAD, magazine and release architecture, subsystem models and system mass rollup.

Analysis delivered

Shot, control, circuit, structure, thermal, orbit and alternative-case results.

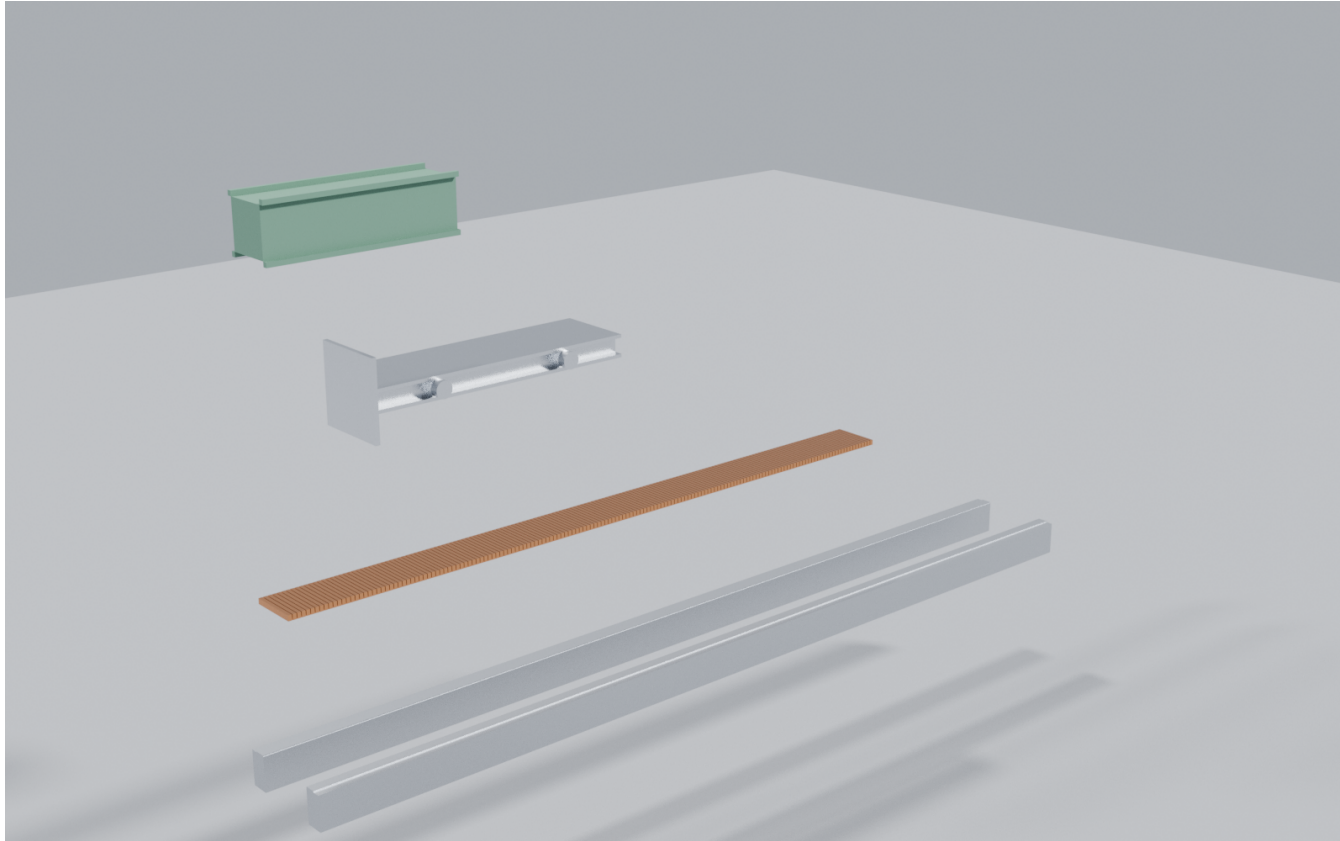
Evidence delivered

Manuscript, figures, scripts, numerical checks, provenance and defect register.

Scope: fixed computational study with failed design gates; no physical qualification claim.

Gen5: fixed reference geometry, feeder fit still open

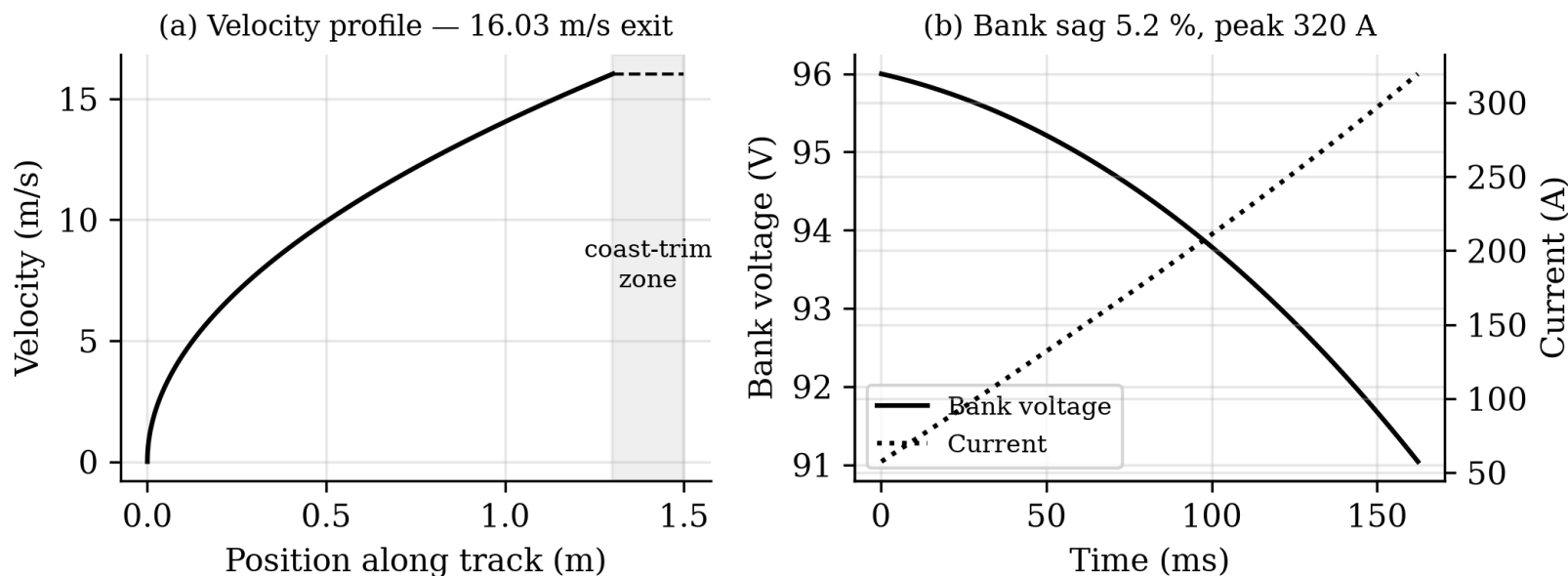
RESULTS



- 1.5 m linear-motor track and double-sided Halbach arrays
- Reusable 9.45 kg magnet sled
- Two cassettes; twelve 3U satellites
- FreeCAD STEP assembly: side-fed placement clashes

Historical periodic-model shot: challenged by finite geometry

RESULTS



16.0 m/s

exit speed

10.07 g

acceleration

2.782 kJ

gross draw

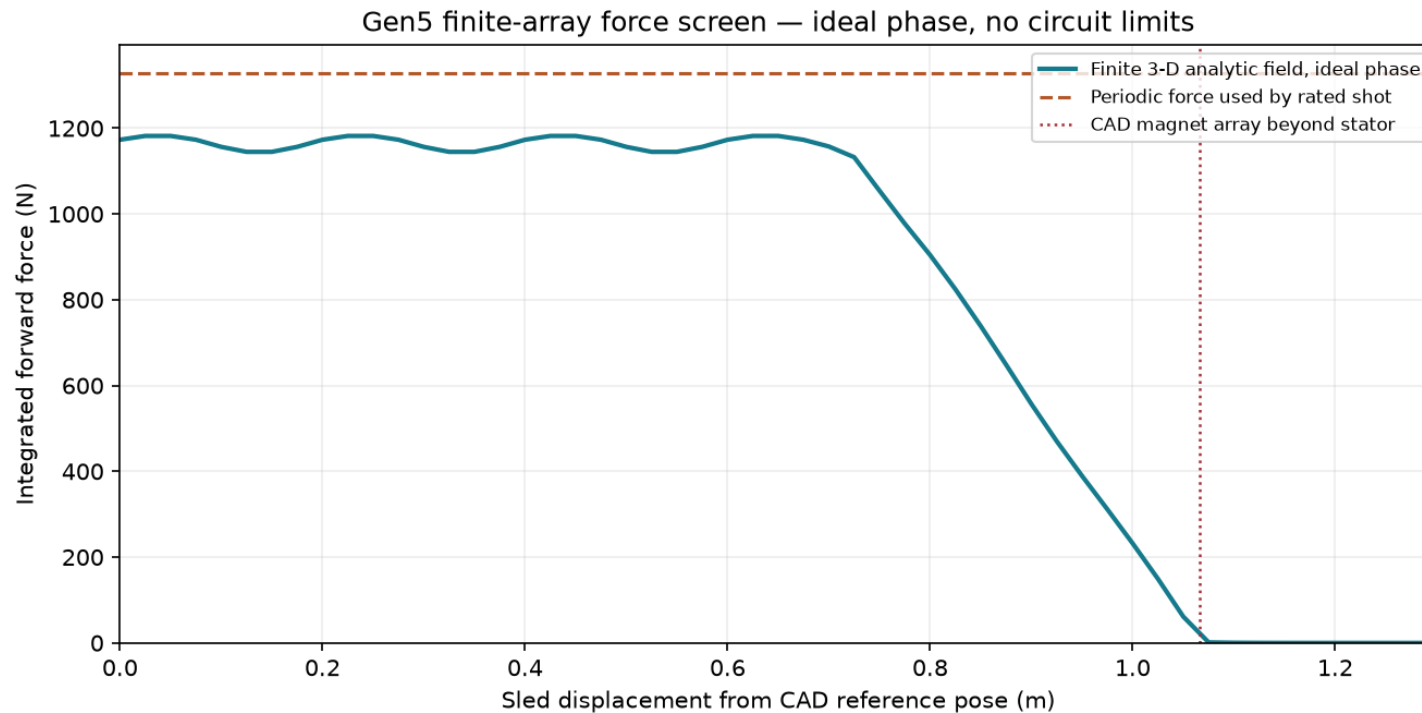
18.8%

net efficiency

P118 finite geometry challenges 16.0 m/s; 124.5 J of gross draw is also unitemized.

Finite-array force integration changes the speed verdict

RESULTS



1.042 kJ

Ideal work from the finite 162-belt force map with optimized phase.

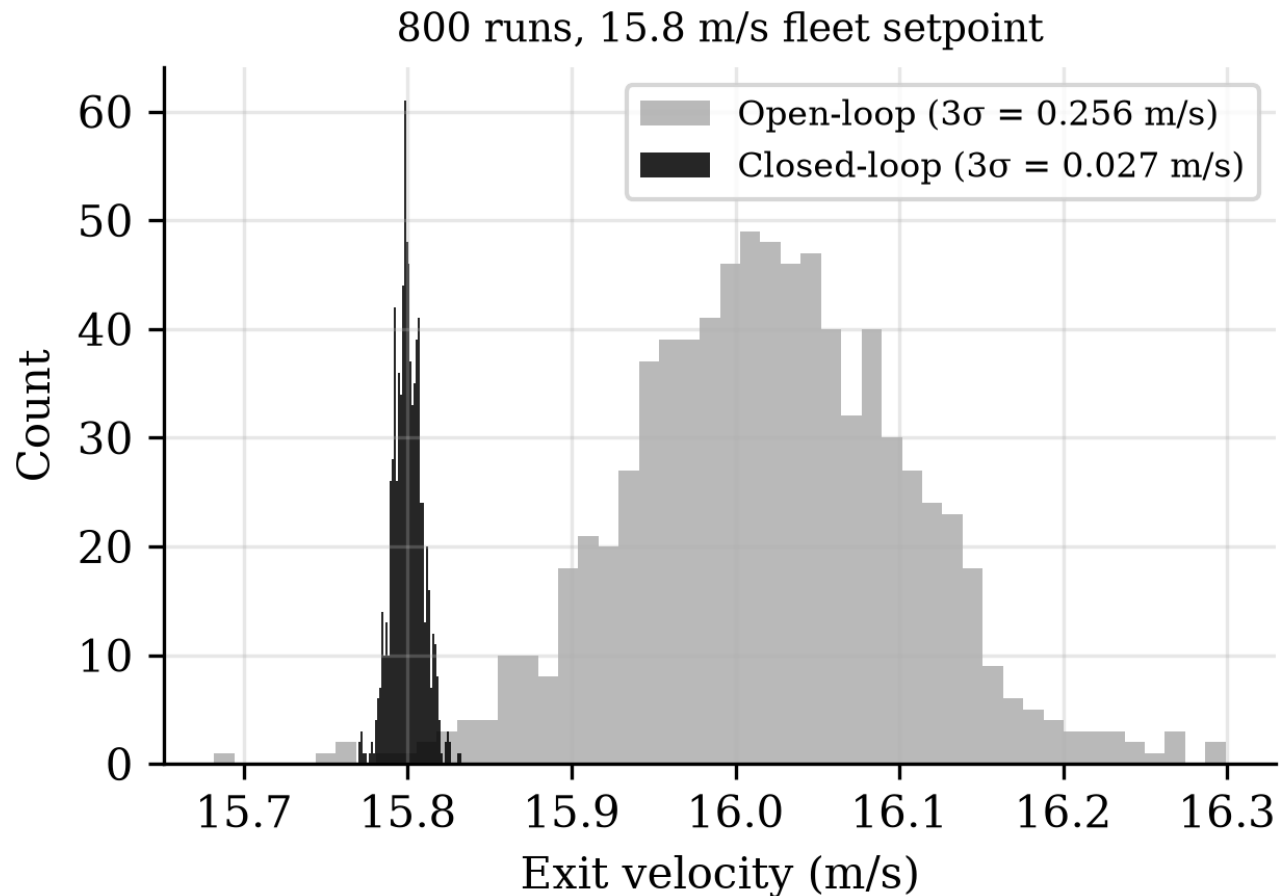
12.448 m/s

Geometry-only speed before circuit and friction losses; historical model reports 16.029 m/s.

Array/stator direct overlap ends after 1.066 m, before the 1.300 m powered stroke ends.

Modelled command precision is narrow; it is unmeasured

RESULTS



0.0274 m/s

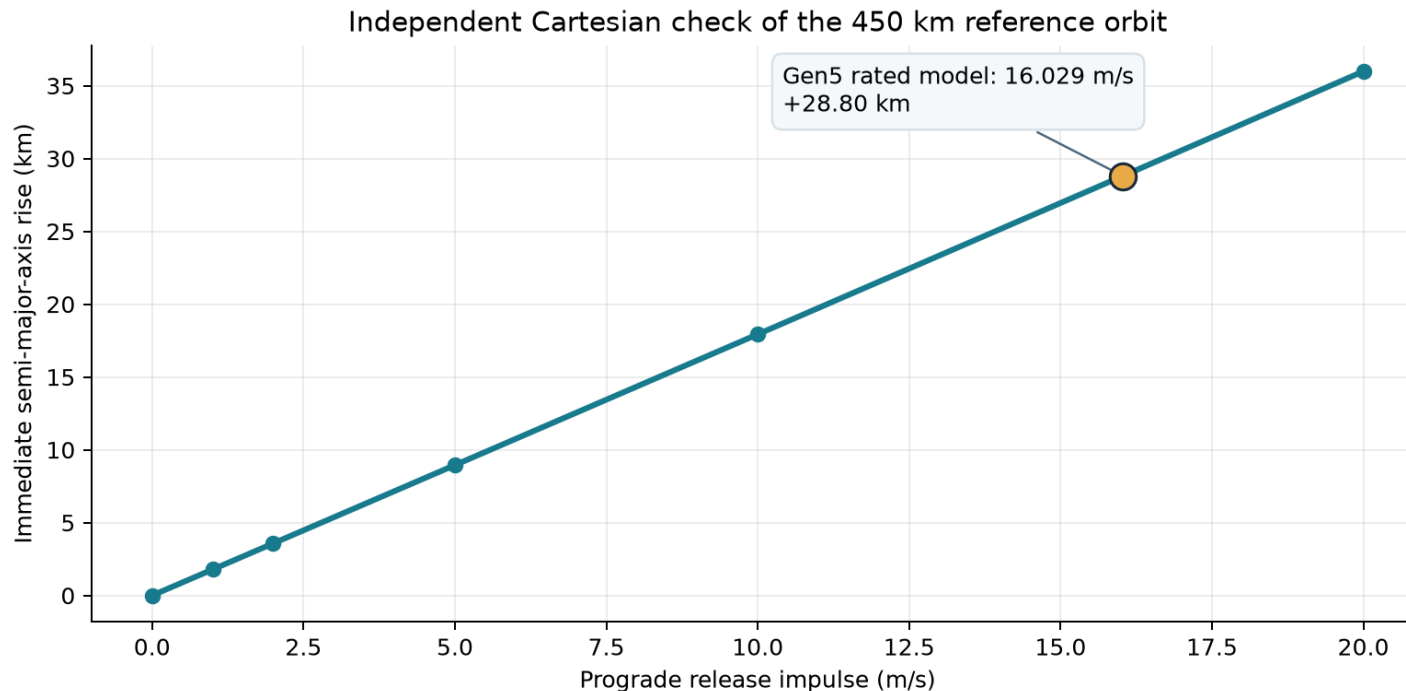
3σ exit-velocity dispersion at a 15.8 m/s fleet setpoint.

Evidence limit

Monte Carlo under modelled sensing and plant tolerances; no measured dispersion.

Historical rated orbit input has a separate geometry check

RESULTS



+28.8008 km

Immediate two-body axis rise
at the rated 16.029 m/s input.

1.60x open

Lifetime remains
atmosphere-dependent without a
current rated independent rerun.

Persistent phasing requires a relative state change; this result concerns orbital energy.

Gen5 fails its 3U mass comparison

RESULTS

Modeled dry mass, 12 slots **126.6 kg**

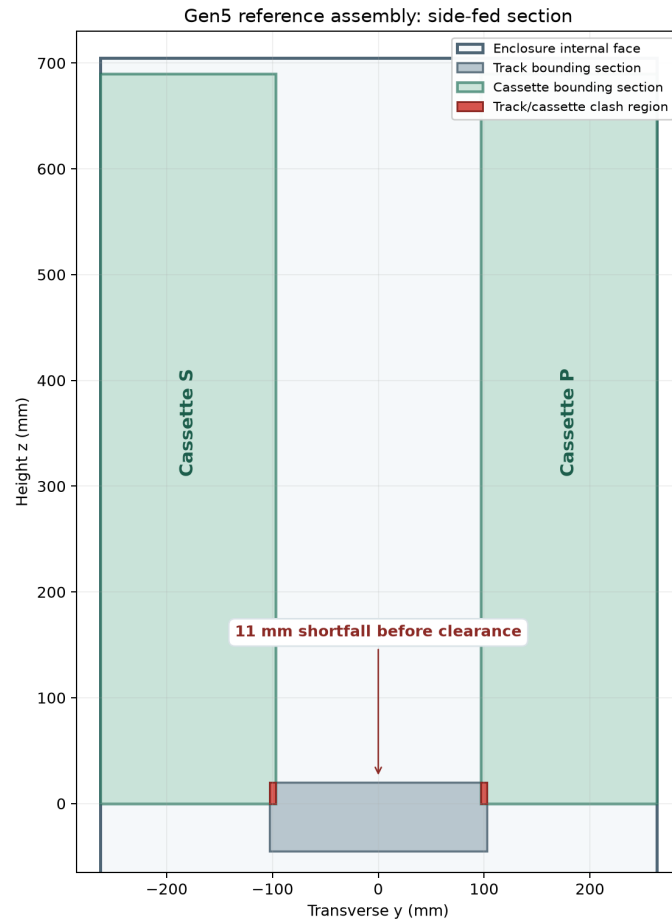
Gen5 per 3U  **10.55 kg**

Spring comparator  **~6 kg**

1.76× heavier per 3U payload; the preset 15% parity band fails.

The current side-fed assembly fails a geometric fit check

RESULTS



Reference placement only: a new feeder or enclosure would need a new configuration and rerun.

11 mm short

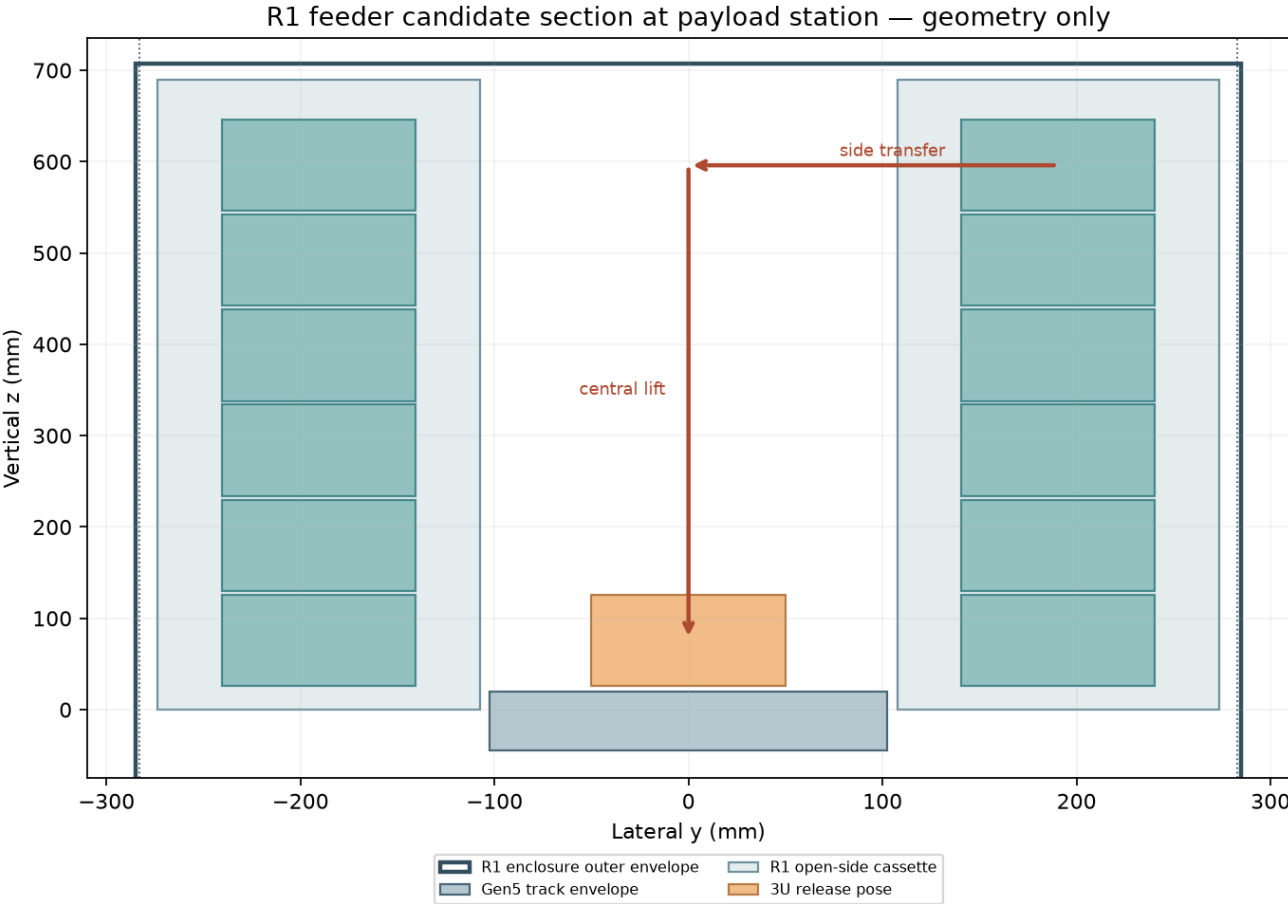
537 mm of track and cassette width inside
526 mm of usable enclosure width.

32,915 mm³

Exact track/cassette clash on each side;
CadQuery and FreeCAD agree.

R1 clears a scripted path; the feeder is not designed

RESULTS



12 / 12

Scripted 3U envelope routes and conservative fixed-part swept boxes clear.

570 mm

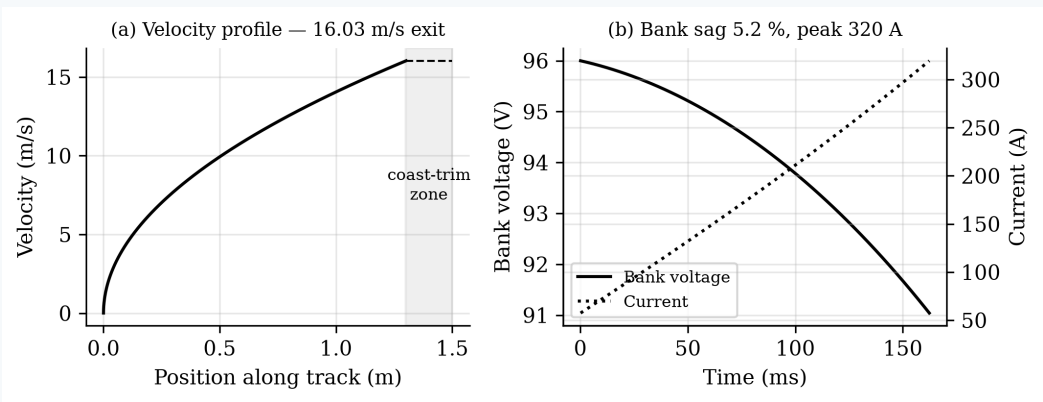
Wider enclosure; lift actuator, retention, tolerance and new mass budget open.

Separate R1 candidate; no change to evaluated Gen5 shot, mass or host-fit claims.

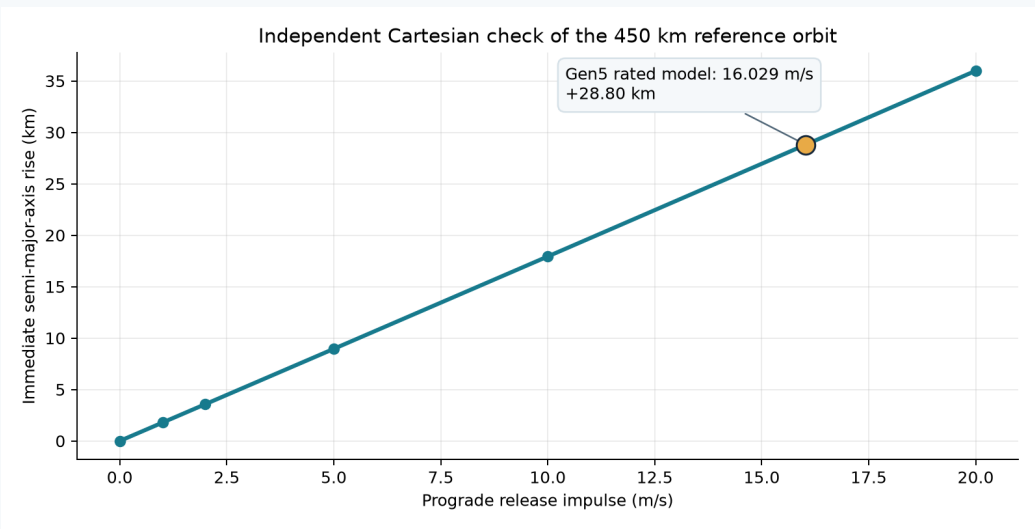
Computational demonstration: from shot to orbit

RESULTS

1. Simulate a 3U shot → 16.0 m/s



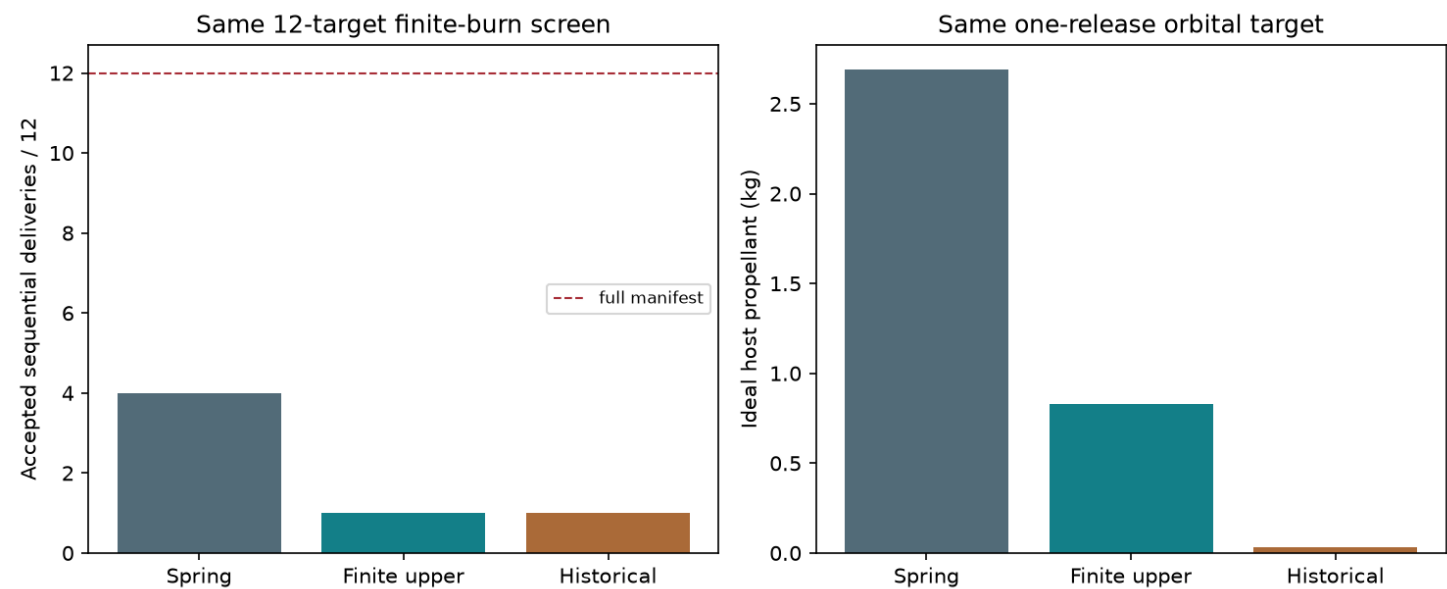
2. Model the orbital response



**Compare predicted orbital change and full installed mass with the spring baseline.
Simulation, not physical demonstration.**

Matched reference mission finds no twelve-shot closure

RESULTS



2.69 vs 0.83 kg

Ideal host propellant for one matched inertial target: spring versus finite Gen5.

4 / 1 / 1

Accepted twelve-shot prefixes: spring, finite Gen5, historical Gen5.

Assumed host and devices; optimizer result is not an infeasibility proof or product advantage.

The research question has a completed answer

CONCLUSION

- Historical model reports 16.0 m/s; a finite force screen challenges that performance claim.
- The modeled system is 76% heavier per 3U customer and the evaluated side-fed CAD clashes.
- R1 geometry clears a scripted path, but has no lift or launch retention design.
- No matched twelve-shot reference closes; Gen5 is a documented negative design study.

Gen6 reopens the mechanism and installed-system trade

FUTURE WORK

One shared path • sequential feed • commanded release for each CubeSat

- Investigate approximately 1–2 m/s through 1 km/s across mission cases.
- Compare electromagnetic, stored-energy, gas/fluid, hybrid, BOLLEY and conventional options.
- Count feeder, retention, arrest, control, energy, thermal, host loads and shared faults.

1 km/s is a research target, not demonstrated speed or payload/provider compatibility.

Next milestone: one reviewable first-test configuration

FUTURE WORK

Freeze

Named mission, payload and host surrogate

Decide

Fair mechanism and installed-burden trade

Measure

Field, contact, release, arrest, loads and cycle behavior

Review

Drawings, BOM, procedures, uncertainty and review decision

Core literature and standards (1/2)

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- [2] Cal Poly, CubeSat Design Specification, Rev. 14, 2020.
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Methods and project records (2/2)

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